

# Artificial Intelligence for Robotics 2

**Assignment D3-V5: Warehouse Robotics– Priority  
Deliveries and Deadlines**

---

**Student Name:** Andrea Laura

**Course:** Artificial Intelligence for Robotics 2

**Academic Year:** 2025–2026

---

**University of Genoa**

May 30, 2026

# 1 Scenario

This project addresses a warehouse logistics scenario in which autonomous robots are responsible for delivering packages to customers. Some packages are more urgent than others and therefore require priority treatment. In addition, deliveries must be completed within specific deadlines.

The scenario reflects real-world logistics environments where urgent shipments, limited resources, and time-sensitive operations require intelligent scheduling decisions.

## 2 Domain

The domain consists of a warehouse containing multiple packages and a fleet of autonomous delivery robots.

The main elements of the domain are:

- **Robots:** one or multiple autonomous robots operating simultaneously.
- **Packages:** multiple packages in the warehouse with different levels of urgency.
- **Locations:** customer locations representing delivery destinations.
- **Constraints:** delivery and return operation, scheduling constraint based on priority and deadlines.

The objective is to generate delivery plans that satisfy all operational requirements while minimizing execution cost.

## 3 Objective

The goal of the project is to develop an Artificial Intelligence planning model capable of:

- Delivering all packages successfully.
- Respecting package priorities.
- Satisfying delivery deadlines.
- Coordinating multiple robots efficiently.
- Minimizing operational cost.

To achieve this objective, two different planning approaches were implemented and compared: Classical PDDL and PDDL+.

## 4 Model Design

Two planning models were developed to represent the same logistics scenario.

### Basic PDDL

The first model was implemented using Classical PDDL. Since Classical PDDL does not directly support time and deadlines, package urgency was represented through ordering constraints. Packages are assigned priority levels and precedence relations force urgent deliveries to be completed before less urgent ones.

### PDDL+

The second model was implemented using PDDL+. This extension allows explicit modelling of time through processes and events. Delivery durations, return durations, priorities and deadlines are represented directly, enabling the planner to reason about schedule feasibility and deadline satisfaction.

## 5 Q1 - Basic PDDL Model

### 5.1 Domain

The Classical PDDL domain models a delivery system composed of robots, packages and locations.

The main predicates represent:

- Robot position (*at*)
- Package position (*package-at*)
- Robot availability (*free*)
- Package status (*delivered*)
- Package destination (*destination*)
- Connectivity between locations (*connected*)

Priority levels were introduced using dedicated predicates:

```
(high-priority ?p - package)
(medium-priority ?p - package)
(low-priority ?p - package)
```

However, the operational priority mechanism is enforced through the predicate:

```
(must-before ?urgent - package ?later - package)
```

This mechanism guarantees that urgent deliveries are completed before less urgent ones. A package can only be picked up or delivered if all packages that must precede it have already been delivered.

The classical model includes three main **actions**:

- **Move**: robots move between connected locations while increasing the total execution cost;
- **Pick-up**: robots collect packages from the warehouse;
- **Deliver**: robots deliver packages to their destinations and update the system state.

Action costs were introduced to encourage efficient plans and allow plan quality optimization.

## 5.2 Problem With NO Urgency or Deadlines

The first problem instance represents a baseline scenario with:

- 2 robots
- 4 packages
- 4 customer locations

No priority constraints are included and no **must-before** predicates are defined.

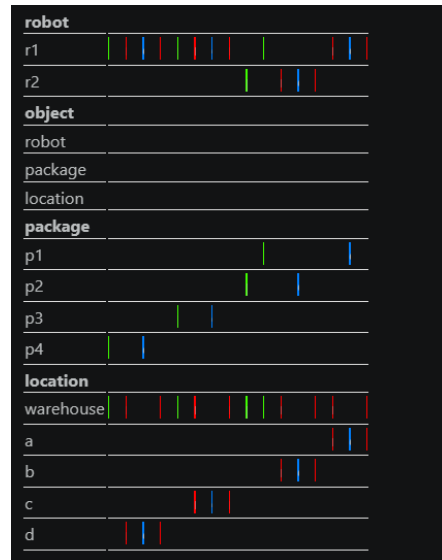
As a result, the planner is free to choose any delivery order, focuses only on minimizing the total cost and both robots return to the warehouse.

```

|pick-up r1 p4 warehouse
|move r1 warehouse d
|deliver r1 p4 d
|move r1 d warehouse
|pick-up r1 p3 warehouse
|move r1 warehouse c
|deliver r1 p3 c
|move r1 c warehouse
|pick-up r2 p2 warehouse
|pick-up r1 p1 warehouse
|move r2 warehouse b
|deliver r2 p2 b
|move r2 b warehouse
|move r1 warehouse a
|deliver r1 p1 a
|move r1 a warehouse

```

(a) Plan found for the problem No-Urgency



(b) Robot utilization

Figure 1: Results for the baseline scenario without urgency constraints

### 5.3 Problem With Urgency or Deadlines

The second problem introduces package priorities. Packages are classified as follows:

| Package | Priority |
|---------|----------|
| p1      | High     |
| p2      | Medium   |
| p3      | Low      |
| p4      | Low      |
| p5      | High     |

Priority levels are enforced through precedence constraints such as:

```

must-before p1 p2
must-before p5 p2
must-before p2 p3
must-before p2 p4

```

This forces high-priority packages to be delivered before medium-priority packages, and medium-priority packages before low-priority ones.

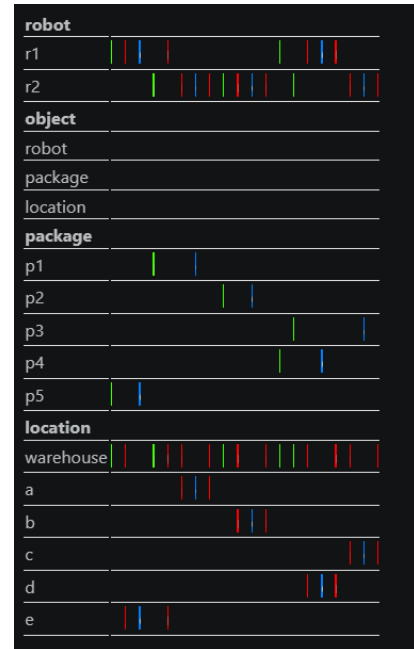
The generated plan respects all precedence constraints while minimizing the total cost, demonstrating that package urgency directly affects delivery ordering and both robots return to the warehouse upon completion of their tasks.

```

|pick-up r1 p5 warehouse
|move r1 warehouse e
|deliver r1 p5 e
|pick-up r2 p1 warehouse
|move r1 e warehouse
|move r2 warehouse a
|deliver r2 p1 a
|move r2 a warehouse
|pick-up r2 p2 warehouse
|move r2 warehouse b
|deliver r2 p2 b
|move r2 b warehouse
|pick-up r1 p4 warehouse
|pick-up r2 p3 warehouse
|move r1 warehouse d
|deliver r1 p4 d
|move r1 d warehouse
|move r2 warehouse c
|deliver r2 p3 c
|move r2 c warehouse

```

(a) Plan found for the problem with Urgency



(b) Robot utilization

Figure 2: Results for the scenario with priority constraints

## 6 Q2 - PDDL+ Model

### 6.1 Domain

The PDDL+ model extends the previous approach by introducing explicit temporal reasoning.

A continuous process called **time-passing** models the progression of time and the accumulation of operational cost.

Package priorities are represented numerically through a priority function. Higher-priority packages must leave the warehouse before lower-priority packages can be assigned.

The model introduces:

- Delivery durations
- Return durations
- Package deadlines
- Continuous delivery processes
- Continuous return processes
- Automatic completion events
- Deadline violation events

Deliveries are executed through a process that continuously increases delivery progress until the required delivery time is reached. Completion is handled automatically by an event.

Similarly, robots return to the warehouse through dedicated return processes and events.

A deadline violation event automatically marks a package as violated whenever its deadline is exceeded before delivery.

### 6.2 Problem

The PDDL+ Problem contains :

- 2 Robots
- 5 Packages and 5 Customer locations
- Delivery durations
- Return durations
- Deadlines
- Priority values

The configuration of the problem is shown below:

| Package | Priority | Cost time | Deadline | Return cost |
|---------|----------|-----------|----------|-------------|
| p1      | 10       | 5         | 6        | 2           |
| p2      | 8        | 5         | 10       | 2           |
| p5      | 6        | 5         | 12       | 2           |
| p3      | 4        | 5         | 16       | 2           |
| p4      | 2        | 5         | 20       | 2           |

The goal requires all packages to be delivered, no deadline violations, and both robots returned to the warehouse.

## 6.3 Results

### 6.3.1 Valid Schedule

The planner successfully generated valid temporal schedules where both robots operate concurrently while respecting all deadlines.

This demonstrates the effectiveness of temporal planning and multi-robot coordination.

```

found plan:
0: (start-delivery r1 p1 warehouse a)
0: (start-delivery r2 p2 warehouse b)
0: -----waiting---- [5.0]
5.0: (start-return r2 b warehouse)
5.0: (start-return r1 a warehouse)
5.0: -----waiting---- [7.0]
7.0: (start-delivery r1 p5 warehouse e)
7.0: (start-delivery r2 p3 warehouse c)
7.0: -----waiting---- [12.0]
12.0: (start-return r2 c warehouse)
12.0: (start-return r1 e warehouse)
12.0: -----waiting---- [14.0]
14.0: (start-delivery r2 p4 warehouse d)
14.0: -----waiting---- [19.0]
19.0: (start-return r2 d warehouse)
19.0: -----waiting---- [21.0]

```

Figure 3: PDDL+ Plan generated for Problem with priority

### 6.3.2 Deadline Violation Experiment

To test the deadline mechanism, some deadlines were intentionally reduced.

When a package could no longer be delivered before its deadline, the planner returned: `problem unsolvable`

This behaviour is expected because the deadline-violation event automatically marks late packages as violated. Since the goal requires all packages to be delivered without violations, the planner correctly determines that no feasible solution exists.

This experiment demonstrates that scheduling decisions directly affect plan feasibility.

```
domain parsed
problem parsed
grounding..
grounding time: 52
aibr preprocessing
|f|:49
|x|:8
heuristic problem creation
aibr preprocessing
|a|:20
|p|:21
|e|:25
delta time heuristic model:1.0
delta time planning model:1.0
delta time search-execution model:1.0
delta time validation model:1
h1 setup time (msec): 12
problem unsolvable
metric (search):-1.0
planning time (msec): 48
heuristic time (msec): 23
```

Figure 4: Plan unsolvable because of deadline violation

## 7 Conclusion

This project compared Classical PDDL and PDDL+ for warehouse delivery planning.

The Classical PDDL model successfully represents package urgency through precedence constraints and generates valid delivery sequences.

The PDDL+ model extends this approach by introducing explicit temporal reasoning, delivery durations and deadline management. This allows the planner to reason directly about schedule feasibility and the consequences of poor scheduling decisions.

Overall, the results show that PDDL+ provides a more realistic representation of warehouse logistics and is particularly suitable for problems involving deadlines and temporal constraints.

Although the problem instances presented in this project are intentionally simple, the models can be easily extended to represent more complex and realistic warehouse environments. For example, additional robots, packages, storage areas, customer locations, or more detailed warehouse layouts could be introduced to increase the complexity of the planning problem and further evaluate the scalability of the proposed approaches.